

4.8 L'Hopital's Rule; Indeterminate Forms Worksheet

1. Find $\lim_{x \rightarrow 0} \left(\frac{3x - \sin(x)}{x} \right)$.

2. Find $\lim_{x \rightarrow \infty} \frac{e^x}{x^2}$.

3. Find $\lim_{x \rightarrow 0} \left(\frac{\sqrt{1+x} - 1 - \frac{x}{2}}{x^2} \right)$.

4. Find $\lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{1}{\arctan(x)} \right)$

5. Find $\lim_{x \rightarrow 0} \left(\frac{1 - \cos(x)}{x + x^2} \right)$.

6. Find $\lim_{x \rightarrow \infty} \left(x \tan \left(\frac{1}{x} \right) \right)$.

7. Find $\lim_{x \rightarrow \infty} (e^x + x)^{\frac{1}{x}}$.

8. Find $\lim_{x \rightarrow 0^+} x^{\sin(x)}$.